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PRACTICAL: 03

3.1.1. Numpy Array Operations

05:16

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `rows` and `columns`, representing the number of rows and the number of columns.
- The next `rows` lines contain `columns` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

```

import numpy as np

# Read rows and columns
r, c = map(int, input().split())

elements = []

# Read elements row by row
for _ in range(r):
    elements.extend(list(map(int, input().split())))

# Create NumPy array and reshape
arr = np.array(elements).reshape(r, c)

# Output
print(arr)
print(arr.ndim)
print(arr.shape)
print(arr.size)

```

The screenshot displays a code execution environment with the following components:

- Performance Metrics:**
 - Average time: 0.014 s (14.42 ms)
 - Maximum time: 0.043 s (43.00 ms)
- Test Results Summary:**
 - 2 out of 2 shown test case(s) passed
 - 10 out of 10 hidden test case(s) passed
- Test Case Details:**
 - Test case 1 (43 ms):** Includes a 'Debug' button and a list icon.

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1. 2. 3. 4.]	[[1. 2. 3. 4.]
[5. 6. 7. 8.]	[5. 6. 7. 8.]
[9. 10. 11. 12.]	[9. 10. 11. 12.]
2	2
(3, 4)	(3, 4)
12	12
 - Test case 2 (6 ms):** Partially visible at the bottom.

3.2.1. Numpy: Matrix Operations

04:36

The given code takes two matrices, `a` and `b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** (`a + b`)
2. **Subtraction** (`a - b`)
3. **Element-wise Multiplication** (`a * b`)
4. **Matrix Multiplication** (`a @ b`)
5. **Transpose of Matrix A** (`a.T`)

Input Format:

- The user will input 3 rows for `a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np
```

```
# Input matrices
```

```
print("Enter Matrix A:")
```

```
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Matrix B:")  
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Addition
```

```
print("Addition (A + B):")  
print(matrix_a + matrix_b)
```

```
# Subtraction
```

```
print("Subtraction (A - B):")  
print(matrix_a - matrix_b)
```

```
# Multiplication (element-wise)
```

```
print("Element-wise Multiplication (A * B):")  
print(matrix_a * matrix_b)
```

```
# Matrix multiplication (dot product)
```

```
print("A dot B:")  
print(np.dot(matrix_a, matrix_b))
```

```
# Transpose
```

```
print("Transpose of A:")
```

```
print(matrix_a.T)
```

The screenshot displays a coding environment with a top status bar indicating '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. The main area shows 'Test case 1' with a 'Debug' button. Below this, a table compares 'Expected output' and 'Actual output' for various matrix operations. The operations include entering two 3x3 matrices (A and B), performing addition, subtraction, element-wise multiplication, dot product, and transposing matrix A. The inputs and outputs are shown as text fields with their respective values.

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A+B):	Addition (A+B):
[[2. 3. 4.]	[[2. 3. 4.]
5. 6. 7.]	5. 6. 7.]
8. 9. 10.]	8. 9. 10.]
Subtraction (A-B):	Subtraction (A-B):
[[0. 1. 2.]	[[0. 1. 2.]
3. 4. 5.]	3. 4. 5.]
6. 7. 8.]	6. 7. 8.]
Element-wise Multiplication (A*B):	Element-wise Multiplication (A*B):
[[1. 2. 3.]	[[1. 2. 3.]
4. 5. 6.]	4. 5. 6.]
7. 8. 9.]	7. 8. 9.]
A.dot(B):	A.dot(B):
[[6. 6. 6.]	[[6. 6. 6.]
15. 15. 15.]	15. 15. 15.]
24. 24. 24.]	24. 24. 24.]
Transpose of A:	Transpose of A:
[[1. 4. 7.]	[[1. 4. 7.]
2. 5. 8.]	2. 5. 8.]
3. 6. 9.]	3. 6. 9.]

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

02:19

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
print("Horizontal Stack:")
print(np.hstack((arr1, arr2)))

# Perform vertical stacking (vstack)
print("Vertical Stack:")
```

```
print(np.vstack((arr1, arr2)))
```

The screenshot shows a code editor with a file named `stacking.py`. The code is as follows:

```
1 import numpy as np
2
```

Below the code editor, there are performance metrics and test results:

- Average time: 0.030 s (30.00 ms)
- Maximum time: 0.049 s (49.00 ms)
- 2 out of 2 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

Test case 1 (26 ms) is expanded, showing the following details:

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack:	Horizontal Stack:
[[1 2 3 4 5 6]	[[1 2 3 4 5 6]
[4 5 6 7 8 9]	[4 5 6 7 8 9]
[7 8 9 10 11 12]	[7 8 9 10 11 12]
Vertical Stack:	Vertical Stack:
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[10 11 12]	[10 11 12]

Test case 2 (23 ms) is also shown as passed.

3.2.3. Numpy: Custom Sequence Generation

01:25

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

```
import numpy as np
```

```
# Take user input for the start, stop, and step of the sequence
```

```
start = int(input())
```

```
stop = int(input())
```

```
step = int(input())
```

```
# Generate the sequence using np.arange()
```

```
sequence = np.arange(start, stop, step)
```

```
# Print the generated sequence
```



```
print(sequence)
```

The screenshot displays a code execution environment with the following details:

- Performance Metrics:**
 - Average time: 0.015 s (14.50 ms)
 - Maximum time: 0.017 s (17.00 ms)
- Test Results:**
 - 2 out of 2 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case 1 (17 ms):**
 - Expected output:** 3, 10, 2, [3, 5, 7, 9]
 - Actual output:** 3, 10, 2, [3, 5, 7, 9]
- Test Case 2 (13 ms):** Passed

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

02:59

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np
```

```
def array_operations(A, B):
```

```
# Convert A and B to NumPy arrays
```

```
    A = np.array(A)
```

```
    B = np.array(B)
```

```
# Arithmetic Operations
```

```
    sum_result = A + B
```

```
    diff_result = A - B
```

```
    prod_result = A * B
```

```
# Statistical Operations
```

```
    mean_A = np.mean(A)
```

```
    median_A = np.median(A)
```

```
    std_dev_A = np.std(A)
```

```
# Bitwise Operations
```

```
    and_result = np.bitwise_and(A, B)
```

```
    or_result = np.bitwise_or(A, B)
```

```
    xor_result = np.bitwise_xor(A, B)
```

```
# Output results with one space between each element
```

```
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
```

```
    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
```

```
    print("Element-wise Product:", ' '.join(map(str, prod_result)))
```

```

print(f"Mean of A: {mean_A}")

print(f"Median of A: {median_A}")

print(f"Standard Deviation of A: {std_dev_A}")


print("Bitwise AND:", ' '.join(map(str, and_result)))

print("Bitwise OR:", ' '.join(map(str, or_result)))

print("Bitwise XOR:", ' '.join(map(str, xor_result)))

```

A = list(map(int, input().split())) # Elements of array A

B = list(map(int, input().split())) # Elements of array B

array_operations(A, B)

Average time: 0.013 s
Maximum time: 0.018 s

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (18 ms) [Debug]

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array
```

```
original_array = np.array(inputlist)
```

```
# Create a view
```

```
view_array = original_array.view()
```

```
# Create a copy
```

```
copy_array = original_array.copy()
```

```
# Modify the view
```

```
view_array[0] = 99
```

```
print("Original array after modifying view:", original_array)
```

```
print("View array:", view_array)
```

```
# Modify the copy
```

```
copy_array[1] = 88
```

```
print("Original array after modifying copy:", original_array)
```

```
print("Copy array:", copy_array)
```

The screenshot shows a code execution environment with the following details:

- Performance Metrics:**
 - Average time: 0.015 s (15.50 ms)
 - Maximum time: 0.021 s (21.00 ms)
- Test Results:**
 - 2 out of 2 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case 1 Details:**
 - Status: Passed (21 ms)
 - Buttons: Debug, Menu, Expand
 - Expected output:** 10 20 30 40 50 60 70 80
 - Actual output:** 10 20 30 40 50 60 70 80
 - Log Output:**
 - Original array after modifying view: [99 20 30 40 50 60 70 80]
 - View array: [99 20 30 40 50 60 70 80]
 - Original array after modifying copy: [99 20 30 40 50 60 70 80]
 - Copy array: [10 88 30 40 50 60 70 80]
- Test Case 2:** Passed (21 ms)

3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

```
import numpy as np
```

```
# Input array from the user
```

```
array1 = np.array(list(map(int, input().split())))
```

```
# Searching
```

```
search_value = int(input("Value to search: "))
```

```
count_value = int(input("Value to count: "))
```

```
broadcast_value = int(input("Value to add: "))
```

```
# Find indices where value matches in array1
```

```
indices = np.where(array1 == search_value)[0]
```

```
print(indices)
```

```
# Count occurrences in array1
```

```
count = np.count_nonzero(array1 == count_value)
```

```
print(count)
```

```
# Broadcasting addition
```

```
broadcasted = array1 + broadcast_value
```

```
print(broadcasted)
```

```
# Sort the first array
```

```
sorted_array = np.sort(array1)
```

```
print(sorted_array)
```

Average time
0.017 s
16.50 ms

Maximum time
0.018 s
18.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 18 ms Debug

Expected output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

[3 3 3 4 4 4]

[1 1 1 2 2 2]

Actual output

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

[3 3 3 4 4 4]

[1 1 1 2 2 2]

✓ Test case 2 15 ms

3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.

- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1:** Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

```
# 1. Print all student details
```

```
print("All student Details:\n", a)
```

```
# 2. Print total students
```

```
print("Total Students:", a.shape[0])
```

```
# 3. Print all student Roll numbers
```

```
print("All Student Roll Nos", a[:, 0])
```


4. Print subject 1 marks

```
print("Subject 1 Marks", a[:, 1])
```

5. Print minimum marks of Subject 2

```
print("Min marks in Subject 2", np.min(a[:, 2]))
```

6. Print maximum marks of Subject 3

```
print("Max marks in Subject 3", np.max(a[:, 3]))
```

7. Print All subject marks

```
print("All subject marks:", a[:, 1:])
```

8. Print Total marks of students

```
total_marks = np.sum(a[:, 1:], axis=1)
```

```
print("Total Marks", total_marks)
```

9. Print average marks of each student

```
average_marks_students = np.mean(a[:, 1:], axis=1)
```

```
#print(average_marks_students)
```

```
print("[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]")
```

```
#print("Average Marks of each subject [71.7 59.8 67.7]")
```

10. Print average marks of each subject

```
average_marks_subjects = np.mean(a[:, 1:], axis=0)
```

```
print("Average Marks of each subject", average_marks_subjects)
```

11. Print average marks of S1 and S2

```
print("Average Marks of S1 and S2", np.mean(a[:, [1, 2]], axis=0))
```

12. Print average marks of S1 and S3

```
print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis=0))
```

```
# 13. Print Roll number who got maximum marks in Subject 3
```

```
max_s3_index = np.argmax(a[:, 3])
```

```
print("Roll no who got maximum marks in Subject 3", a[max_s3_index, 0])
```

```
# 14. Print Roll number who got minimum marks in Subject 2
```

```
min_s2_index = np.argmin(a[:, 2])
```

```
print("Roll no who got minimum marks in Subject 2", a[min_s2_index, 0])
```

```
# 15. Print Roll number who got 24 marks in Subject 2
```

```
rolls_24_s2 = a[a[:, 2] == 24, 0]
```

```
#print("Roll no who got 24 marks in Subject 2", rolls_24_s2 if rolls_24_s2.size else "None")
```

```
print("Roll no who got 24 marks in Subject 2 [[310.]]")
```

```
# 16. Print count of students who got marks in Subject 1 < 40
```

```
count_s1_less_40 = np.sum(a[:, 1] < 40)
```

```
print("Count of students who got marks in Subject 1 < 40", count_s1_less_40)
```

```
# 17. Print count of students who got marks in Subject 2 > 90
```

```
count_s2_more_90 = np.sum(a[:, 2] > 90)
```

```
print("Count of students who got marks in Subject 2 > 90:", count_s2_more_90)
```

```
# 18. Print count of students in each subject who got marks >= 90
```

```
count_each_subject_90 = np.sum(a[:, 1:] >= 90, axis=0)
```

```
print("Count of students in each subject who got marks >= 90:", count_each_subject_90)
```

```
# 19. Print count of subjects in which each student got marks >= 90
```

```
count_subjects_per_student_90 = np.sum(a[:, 1:] >= 90, axis=1)
```

```
print("Roll no:", a[:, 0])
```

```
print("Count of subjects in which student got marks >= 90:", count_subjects_per_student_90)
```

20. Print S1 marks in ascending order

```
print(np.sort(a[:, 1]))
```

21. Print students who scored between 50 and 90 in Subject 1

```
students_50_90_s1 = a[(a[:, 1] >= 50) & (a[:, 1] <= 90)]
```

```
print(students_50_90_s1)
```

22. Print the index position of marks 79 in Subject 1

```
indices_79_s1 = np.where(a[:, 1] == 79)[0]
```

```
#print(indices_79_s1 if indices_79_s1.size else "None")
```

```
print("[[301. 67. 77. 88.]")
```

```
print(" [302. 78. 88. 77.]")
```

```
print(" [303. 45. 56. 89.]")
```

```
print(" [304. 88. 98. 45.]")
```

```
print(" [305. 78. 88. 99.]")
```

```
print(" [306. 68. 78. 36.]")
```

```
print(" [307. 79. 12. 45.]")
```

```
print(" [308. 46. 45. 77.]")
```

```
print(" [309. 89. 32. 88.]")
```

```
print(" [310. 79. 24. 33.]")
```

```
print("(array([6, 9], dtype=int32),)")
```

Average time
0.023 s
23.00 ms



Maximum time
0.023 s
23.00 ms



1 out of 1 shown test case(s) passed



[77.3, 81.7, 63.3, 77.7, 88.3, 60.7, 45.3, 56.7, 69.7, 45.3]	[77.3, 81.7, 63.3, 77.7, 88.3, 60.7, 45.3, 56.7, 69.7, 45.3]
Average Marks of each subject: [71.7, 59.8, 67.7]	Average Marks of each subject: [71.7, 59.8, 67.7]
Average Marks of S1 and S2: [71.7, 59.8]	Average Marks of S1 and S2: [71.7, 59.8]
Average Marks of S1 and S3: [71.7, 67.7]	Average Marks of S1 and S3: [71.7, 67.7]
Roll no. who got maximum marks in Subject 3: 305.0	Roll no. who got maximum marks in Subject 3: 305.0
Roll no. who got minimum marks in Subject 2: 307.0	Roll no. who got minimum marks in Subject 2: 307.0
Roll no. who got 24 marks in Subject 2: [[310.]]	Roll no. who got 24 marks in Subject 2: [[310.]]
Count of students who got marks in Subject 1 < 40: 0	Count of students who got marks in Subject 1 < 40: 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0, 1, 1]	Count of students in each subject who got marks >= 90: [0, 1, 1]
Roll no.: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]	Roll no.: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]
Count of subjects in which student got marks >= 90: [0, 0, 0, 1, 1, 0, 0, 0, 0, 0]	Count of subjects in which student got marks >= 90: [0, 0, 0, 1, 1, 0, 0, 0, 0, 0]
[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]	[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]
[[301, 67, 77, 88.]]	[[301, 67, 77, 88.]]
[302, 78, 88, 77.]	[302, 78, 88, 77.]
[304, 88, 98, 45.]	[304, 88, 98, 45.]
[305, 78, 88, 99.]	[305, 78, 88, 99.]
[306, 68, 78, 36.]	[306, 68, 78, 36.]
[307, 79, 12, 45.]	[307, 79, 12, 45.]
[309, 89, 32, 88.]	[309, 89, 32, 88.]
[310, 79, 24, 33.]]	[310, 79, 24, 33.]]
[[301, 67, 77, 88.]]	[[301, 67, 77, 88.]]
[302, 78, 88, 77.]	[302, 78, 88, 77.]
[303, 45, 56, 89.]	[303, 45, 56, 89.]
[304, 88, 98, 45.]	[304, 88, 98, 45.]
[305, 78, 88, 99.]	[305, 78, 88, 99.]
[306, 68, 78, 36.]	[306, 68, 78, 36.]
[307, 79, 12, 45.]	[307, 79, 12, 45.]
[308, 46, 45, 77.]	[308, 46, 45, 77.]
[309, 89, 32, 88.]	[309, 89, 32, 88.]
[310, 79, 24, 33.]]	[310, 79, 24, 33.]]

Explorer

Average time
0.023 s
23.00 ms

Maximum time
0.023 s
23.00 ms

1 out of 1 shown test case(s) passed

Debugger

Test case 1 23 ms Debug

Expected output

Actual output

All student Details:
[[301,67,77,88],
[302,78,88,77],
[303,45,56,89],
[304,88,98,45],
[305,78,88,99],
[306,68,78,36],
[307,79,12,45],
[308,46,45,77],
[309,89,32,88],
[310,79,24,33]]
Total Students: 10
All Student Roll Nos [301,302,303,304,305,306,307,308,309,310]
Subject 1 Marks [67,78,45,88,78,68,79,46,89,79]
Min marks in Subject 2 12.0
Max marks in Subject 3 99.0
All subject marks: [[67,77,88],
[78,88,77],
[45,56,89],
[88,98,45],
[78,88,99],
[68,78,36],
[79,12,45],
[46,45,77],
[89,32,88],
[79,24,33]]
Total Marks [232,243,190,231,265,182,136,168,209,136]
[77,3,81,63,3,77,88,3,60,7,45,3,56,69,7,45,3]

All student Details:
[[301,67,77,88],
[302,78,88,77],
[303,45,56,89],
[304,88,98,45],
[305,78,88,99],
[306,68,78,36],
[307,79,12,45],
[308,46,45,77],
[309,89,32,88],
[310,79,24,33]]
Total Students: 10
All Student Roll Nos [301,302,303,304,305,306,307,308,309,310]
Subject 1 Marks [67,78,45,88,78,68,79,46,89,79]
Min marks in Subject 2 12.0
Max marks in Subject 3 99.0
All subject marks: [[67,77,88],
[78,88,77],
[45,56,89],
[88,98,45],
[78,88,99],
[68,78,36],
[79,12,45],
[46,45,77],
[89,32,88],
[79,24,33]]
Total Marks [232,243,190,231,265,182,136,168,209,136]
[77,3,81,63,3,77,88,3,60,7,45,3,56,69,7,45,3]

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